
西湘研民畫

• Dynamical systems

• Linear differential equations

Examples

$x' = 0$ | $x' - x = 0$

$x' = -2t$ | $x'' - x = 0$

$x' = \sin t$ | $x'' + x = 0$

$x'' = 0$

$x' = 0 \quad x = c, \; c \in \mathbb{R}$

$S = \{c \cdot 1, c \in \mathbb{R}\}$

is a sub of $C(\mathbb{R})$

$x' = -2t \qquad x = t^2 + c, \; c \in \mathbb{R}$

$x'' = 0 \qquad x' = c_1 \qquad x = c_1 t + c_2, \; c_1, c_2 \in \mathbb{R}$

$S = \{c_1 \cdot t + c_2 \cdot 1, \; c_1, c_2 \in \mathbb{R}\}$

$x' - x = 0 \iff x' = x$

$e^t, \; 0 \in \{c \cdot e^t, \; c \in \mathbb{R}\} = \tilde{S}$

Are there other sol.? No

Idea 1 $\quad x' - x = 0 \,|\cdot e^{-t} \qquad x e^{-t} = c \,|\cdot e^t$

$x' e^{-t} - x e^{-t} = 0 \qquad x = c \cdot e^t$

$(x e^{-t})' = 0$

Let S be a set of all solutions of $x' - x = 0$
We know $\tilde{S} \subseteq S$

Idea 2 $x \in \tilde{S} \iff \exists c \in \mathbb{R}, x = c \cdot e^{t} \iff \exists c \in \mathbb{R}$ s.t $x e^{-t} = c$

$x \in S$ $\quad y = x e^{-t}, \ y' = x' e^{-t} - x e^{-t}$ $\Rightarrow y' = 0 \Rightarrow y = c, c \in \mathbb{R}$

$x \in S \Rightarrow x' = x$ $\quad \Rightarrow x e^{-t} = c \Rightarrow x \in \tilde{S}$

Idea 3 S is a subspace of $C(\mathbb{R})$ that (theorem).

"S is a linear space of dim. 1"

$\tilde{S} = \{ c e^{t}, c \in \mathbb{R} \} \subseteq S$

$\tilde{S} = \langle e^{t} \rangle \Rightarrow \tilde{S}$ is a linear space of dim 1

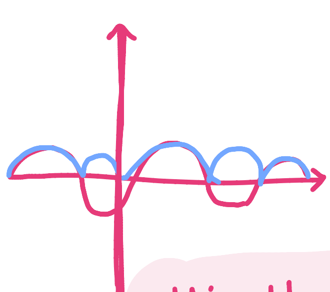
We conclude that $\tilde{S} = S$

$x'' - x = 0$

$0, e^{t}, c \cdot e^{t}$

$x = e^{-t} \Rightarrow x' = -e^{t} \Rightarrow x'' = e^{-t} \Rightarrow x'' = x$

$x = |\sin t|$ $= \begin{cases} \sin t, & t \in I_1 \\ -\sin t, & t \in I_2 \end{cases}$



• $|\sin t|$ graphic

$ch\,t = \frac{1}{2} e^{t} + \frac{1}{2} e^{-t}$ $\quad (ch\,t)' = sh\,t$

$sh\,t = \frac{1}{2} e^{t} - \frac{1}{2} e^{-t}$ $\quad (sh\,t)' = ch\,t$

$ch\,0 = 1 \quad ch\,t^2 - sh\,t^2 = 1$

$sh\,0 = 0$

Are $sh\,t$ and $ch\,t$ a sol of $x'' - x = 0$?

Yes, they are sol.

Theorem:

The set of all solutions of $x'' - x = 0$ in a linear space of dimension 2.

$\Rightarrow$ We need two linearly indep. sol of $x'' - x = 0$

e^{t} $ch\,t$ are solutions.

Are they lin. indep?

Let $\alpha, \beta \in \mathbb{R}$ be s.t $\alpha e^t + \beta \operatorname{ch} t = 0, \forall t \in \mathbb{R}$ it is sufficient to prove that $\alpha = \beta = 0$

$\begin{cases} t = 0 \Rightarrow \alpha + \beta = 0 \\ t = 1 \Rightarrow \alpha e + \beta \operatorname{ch} 1 = 0 \end{cases}$

$$\begin{vmatrix} 1 & 1 \\ e & \operatorname{ch}1 \end{vmatrix} = \operatorname{ch} 1 - e = \frac{1}{2} e + \frac{1}{2} \frac{1}{e} - e = \frac{1}{2}(-e + \frac{1}{e})$$

$$= \frac{1 - e^2}{2e} \neq 0$$

Then $\alpha = \beta = 0$

thus, the 2 funct. are lin indep.

$\beta \neq 0$ $\operatorname{ch} t = -\frac{1}{\beta} e^t$

We have e^t and e^{-t} are lin. indep sol of $x'' - x$.

Also $\operatorname{ch} t$ and $\operatorname{sh} t$ —||— || ——— .

Thus, the gen sol. of $x'' - x = 0$ can also be described by:

$x = c_1 e^t + c_2 e^{-t}, c_1, c_2 \in \mathbb{R}$

or

$x = c_1 \operatorname{ch} t + c_2 \operatorname{sh} t, c_1, c_2 \in \mathbb{R}$

$x'' + x = 0$

Theorem: The set of all sol. is a linear repr. of dim 2.

$\cos t$ and $\sin t$ are sol., lin indep

Thus the $x = c_1 \cos t + c_2 \sin t \in \mathbb{R}$

The general form of an n^{th} order Linear Homogeneous Diff Equ. (LHDE)

(1) $x^{(n)} + a_1(t) x^{(n-1)} + \dots + a_{n-1}(t) x' + a_n(t) x = 0$

then $n \in \mathbb{N}, n \geq 1$ is called the order of the DE (1)

$a_1, a_n \in C(I), I \subset \mathbb{R}$ open interval

they are called coeff.

When $a_1, \dots, a_n$ are constant functions, we say that (1) has constant coefficients.

The Initial Value Problem (IVP) for (1) is formulated

Let $t_0 \in I$ fixed. Let $\eta_1, \dots, \eta_n \in \mathbb{R}$

(IVP) $\begin{cases} eq\ (1) \\ x(t_0)=\eta_1,\ x'(t_0)=\eta_2, \dots, x^{(n-1)}(t_0)=\eta_n \end{cases}$ (intial conditions)

The existence and uniqueness theorem

The (IVP) has a unique solution, $\varphi: I \to \mathbb{R}$

Def A solution of eq (1) is a function $\varphi: I \to \mathbb{R}$ s.t $\varphi \in C^n(I)$ and

$$\varphi^{(n)}(t)+a_1(t)\varphi^{(n-1)}(t)+\dots+a_n(t)\varphi(t)=0, \forall t \in I$$

The foundamental theorem for LHDE

The set of all solution of eq (1) is a linear space of dimension n.

Proof: $C(I)$ has the structure of linear space

$$C^n(I)=\{x: I \to \mathbb{R} : \exists x, x', \dots, x^{(n)}: I \to \mathbb{R} \text{ cont}\}$$

$C^n(I)$ is a subspace of $C(I)$

$x \in C^n(I)$ $\quad \mathcal{L}(x): I \to \mathbb{R}$

$$\mathcal{L}(x)(t)=x^{(n)}(t)+a_1(t)x^{(n-1)}(t)+\dots+a_n(t)x(t), t \in I$$

$$\mathcal{L}: C^n(I) \to C(I)$$

Seminar 1

1.1, 2, 3, 7, 10, 11, 12, 13, 14 → Rest temă

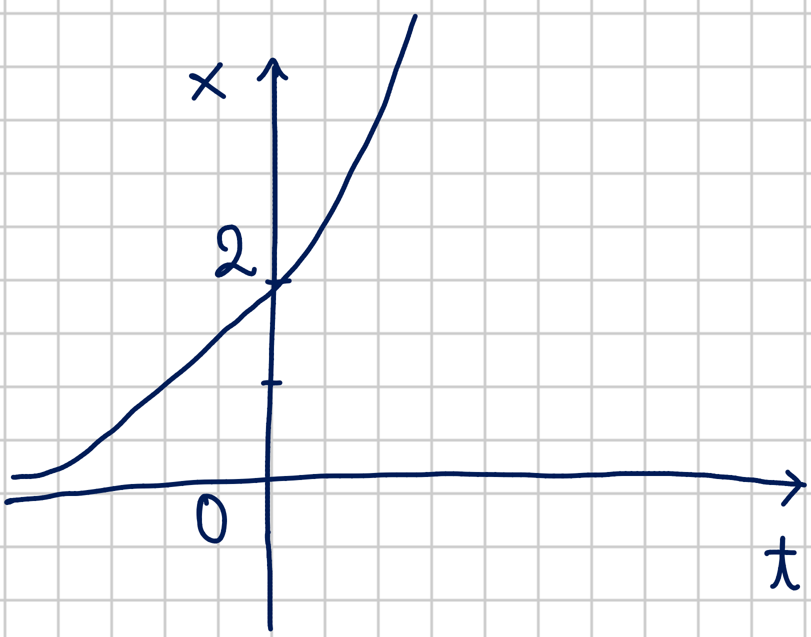
1.1 $f(t) = 2e^{3t}, t \in \mathbb{R}$ $f: \mathbb{R} \to \mathbb{R}$

$x' = 3x$

$x(0) = 2$

$$\begin{cases} f'(t) = 6e^{3t} = 3 \cdot 2e^{3t} = 3f(t), (\forall) t \in \mathbb{R} \\ f(0) = 2e^{3 \cdot 0} = 2 \end{cases}$$

$\Rightarrow f \in S$



Long-term behavior

$\lim_{t \to \infty} f(t) = \infty$ - exponentially increasing

$f(t) > 0, (\forall) t \in \mathbb{R}$

1.2 $f(t) = \frac{1}{6} \sin t$, $t \in \mathbb{R}$

switch ~~to~~' $\frac{1}{6} \to \eta$

$x'' + x = 0$, $x(0) = 0$, $x'(0) = \cancel{\frac{1}{6}}\ \eta$

$f'(t) = \eta \cos t$ $x'' + x = -\eta \sin t + \eta \sin t = 0$

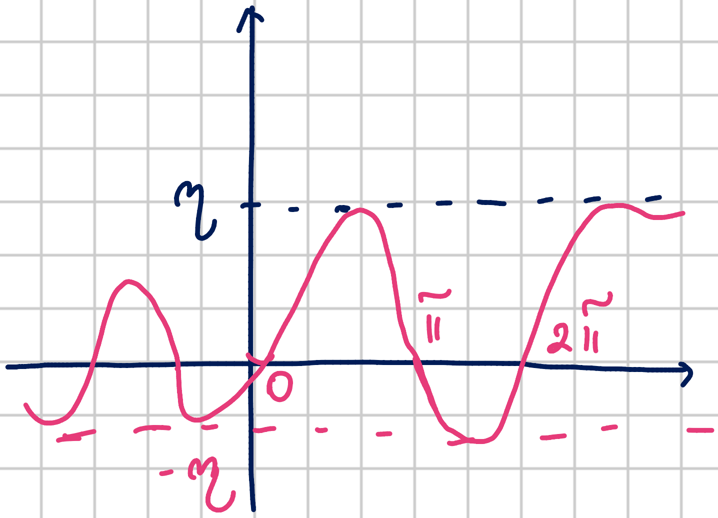
$f''(t) = -\eta \sin t$

$f(0) = \eta \sin(0) = 0$

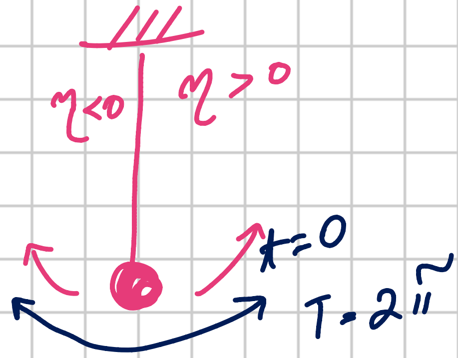
$f'(0) = \eta \cos 0 = \eta$

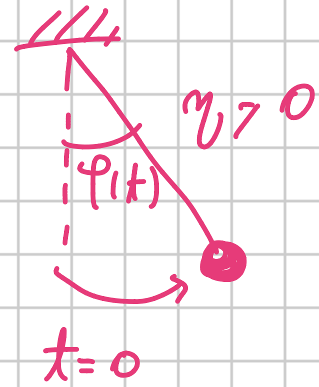
$\Rightarrow f \in S$

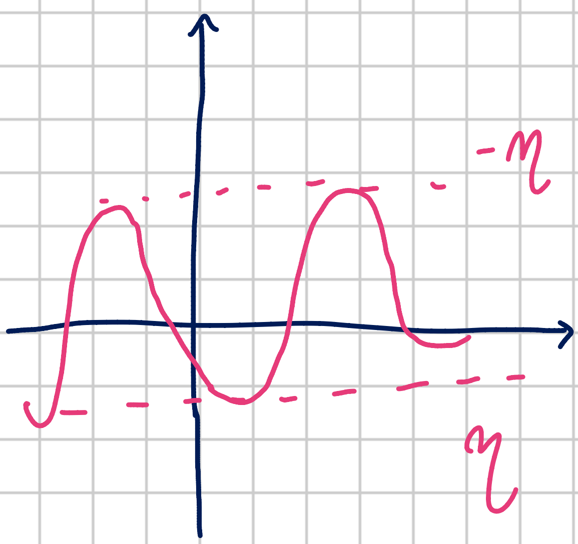
sol for IVP



$\eta > 0$







$\eta < 0$

Long term:

bounded $[-\eta, \eta]$

Periodical function with the main period:

2π

$\eta > 0$

Oscilates around 0 with constant Amptitude (η)

1.3 $f(t) = \frac{1}{6} e^{-2t} \cos t$

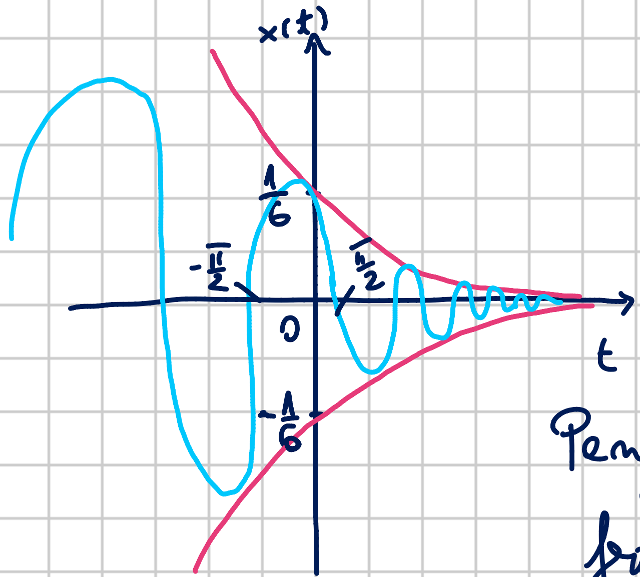
$t \in \mathbb{R}$

$x'' + 4x' + 5x = 0$, $x(0) = \frac{1}{6}$, $x'(0) = -\frac{1}{3}$

$$x' = f'(t) = \frac{1}{6} \cdot (-2) \cdot e^{-2t} \cdot \cos t + \frac{1}{6} \cdot e^{-2t} \cdot (-\sin t)$$

$$= -\frac{1}{3} e^{-2t} \cdot \cos t - \frac{1}{6} \cdot e^{-2t} \sin t$$

$x'' =$



Pendulum with friction

$$\lim_{t\to\infty} \frac{1}{6}\cdot \frac{\overset{\nearrow \text{nmth}}{\cos t}}{\underset{\searrow \infty}{e^{2t}}} \quad 0$$

$$f(t)=0 \iff \frac{1}{6}\cdot\frac{\cos t}{e^{2t}}=0 \iff \cos t = 0$$

$$\Rightarrow t = \underset{\substack{\downarrow \\ \text{period}}}{\frac{\pi}{2}} + k\pi, k\in\mathbb{Z}$$

$\cos t = 1$

$\arccos(\cos(t)) = \arccos(1)$

$\cos t = 2k\pi$

$(2k+1)\pi$

Long term behaviour

- oscilates around 0 with exp. decreasing altitude

$$\lim_{t\to\infty} \frac{1}{6}\,\frac{\cos t}{e^{2t}} = 0$$

Nothing about periodicity (only for cos)

• LECTURE Lecture 2

Linear

Let $n\in\mathbb{N}^*$, $I\subset\mathbb{R}$ open interval $a_1,\dots,a_n\in C(I)$

$C(I), C^1(I), C^n(I)$

$$\mathcal{L}: C^n(I)\to C(I)$$

Exercise

Look for $r\in\mathbb{R}$ s.t. $x=e^{rt}$ is a sol of (2).

$$x' = re^{rt},\ x'' = r^2e^{rt}$$

$$\mathcal{L}(e^{rt}) = e^{rt}\,l(r),\ \forall t\in\mathbb{R}$$

$$l(r) = r^n + a_1r^{n-1} + \dots + a_{n-1}r + a_n \quad (3)$$

e^{rt} is a sol of $\mathcal{L}(x)=0$

$\Leftrightarrow \mathcal{L}(e^{rt})=0, \forall t\in\mathbb{R}$

$\Leftrightarrow \ell(r)=0$, r is a root of the polynomial

Def the polynomial (3) is called the caract. polynomial of the LHDE with CC (2).

while $\ell(r)=0$ is the characteristic equation

Example

Find the caract. pol (eq) of

$x'+ax=0$

$x''+a_1x'+a_2x=0$

$r^2+a_1r+a_2=0$

$x'+ax=0$

$r+a=0$

$r=-a \longmapsto e^{-at}$ a sol

$x\in ce^{at}, c\in\mathbb{R}$

$x''-5x'+6x$

$r^2-5r+6=0$

$r_1=2 \to e^{2t}$

$r_2=3 \to e^{3t}$

$x=c_1e^{2t}+c_2e^{3t}$

$c_1,c_2\in\mathbb{R}$

The roots of a polynomial of degree n (3)

$r^2-2r+1=0$

$(r-1)^2=0$

1 is a double root

ℓ has n roots in $\mathbb{C}$, counted with their corresponding multiplicities.

$r^2+9=0$

$r_1=-3i \quad r_2=3i$

Let $r_1, \dots, r_2 \in \mathbb{C}$ be these roots (not nec. distinct)

Then $\ell(r) = (r-r_1)(r-r_2)\dots\dots(r-r_n)$

Complex solutions of the LHDE (2)

Let $u, v \in C^m(\mathbb{R})$ and $\varphi: \mathbb{R} \to \mathbb{C}$ be $\varphi(t) = u(t) + v(t)$
$t \in \mathbb{R}$

We have $\varphi' = u' + iv'$; $\varphi'' = u'' + iv''$

If $\mathcal{L}(\varphi) = 0$ then φ is a complex sol of $\mathcal{L}(x) = 0$

Prop: If $\varphi: \mathbb{R} \to \mathbb{C}$, $\varphi = u + iv$ (when $u, v: \mathbb{R} \to \mathbb{R}$)
is a complex solution of (2) $\mathcal{L}(x) = 0$
then both u and v are (real) solutions of (2)

Proof $\mathcal{L}(\varphi) = \mathcal{L}(u + iv) = \underbrace{\mathcal{L}(u)}_{real} + i \underbrace{\mathcal{L}(v)}_{real}$

ip $\Rightarrow$ $\mathcal{L}(\varphi) = 0$

$\Rightarrow \mathcal{L}(u) + i\mathcal{L}(v) = 0$

$\Rightarrow \mathcal{L}(u) = \mathcal{L}(v) = 0$

Remark

$r \in \mathbb{C}$ is a root of e

$\Leftrightarrow$ e^{rt} is a (complex) sol. of $L(x)=0$

e^{it} $e^{(1+i)t}$

$r = \alpha + i\beta$ $\alpha, \beta \in \mathbb{R}$

We need the real / part of

$e^{(\alpha i+\beta)t} = u(t) + iv(t)$

Euler's formula

$$e^{\alpha+i\beta} = e^{\alpha} \cdot e^{i\beta} = e^{\alpha}(\cos\beta + i\sin\beta)$$

$t \in \mathbb{R}$

$$e^{(\alpha+i\beta)t} = \underbrace{e^{\alpha t}\cos\beta t}_{} + i\underbrace{e^{\alpha t}\sin\beta t}_{}$$

$\alpha = 0$, $\beta = \pi$ $e^{i\pi} = \cos\pi + i\sin\pi = -1$

$e^{i\pi} = 0$

Lecture 3

Linear diff. equations with constant coeff.

1) LHDE with c.c
2) Ln-HDE with c.c

1) LHDE with c.c

The characteristic equation method

a) $x^{(n)} + x^{(n-1)} a_1 + \dots + a_{n-1} x' + a_n x = 0$

where $n \in \mathbb{N}^*$ (the order of (1)) $a_1, \dots, a_n \in \mathbb{R}$ the coeff of (1))

step 1: Write the characteristic eq of (1)

$$l(r) = r^n + a_1 r^{n-1} + \dots + a_{n-1} r + a_n = 0$$

(2)

step 2: Find $r_1, r_2, \dots r_n \in \mathbb{C}$ the n roots of (2) (note that they can be repeated acording to their multiplicities)

Step 3: We associate a funct. following the rules:

- if $r_1 = \alpha \in \mathbb{R} \to e^{\alpha t}$
- if $r_1 = \alpha + i\beta$ $(\alpha, \beta \in \mathbb{R}, \beta > 0) \mapsto e^{\alpha t}\cos\beta t$, $e^{\alpha t}\sin\beta t$
  simple root

- if $r_1 = \alpha \in \mathbb{R}$ is a root of multiplicity $m \geq 2$ $l \to e^{\alpha t}, te^{\alpha t}, t^2 e^{\alpha t}, \dots, t^{m-1} e^{\alpha t}$

- if $r_{1,2} = \alpha \pm i\beta$ $(\alpha, \beta \in \mathbb{R}, \beta > 0)$ is a root of mult. $m \geq 2$ $\longmapsto e^{\alpha t}\cos\beta t, e^{\alpha t}\sin\beta t, te^{\alpha t}\cos\beta t, \dots, t^{m-1}e^{\alpha t}\sin\beta t$

(2 m functions)

In this way we obtain n functions, let us denote them by:

$$x_1, \dots, x_n$$

Step 4 Write the general sol of (1),

$$x = c_1 x_1 + \dots + c_n x_n \quad , c_1, \dots, c_n \in \mathbb{R}$$

Th. The n funct. $x_1 \dots x_n$ obtained at step 3 are solutions to the LHDE(1) and they are linearly indep.

Notations: $\mathcal{L} : C^n(\mathbb{R}) \to C(\mathbb{R})$

$$\mathcal{L}(x) = x^{(n)} + a_1 x^{(n-1)} + \dots + a_n x$$

So, $(1) \Leftrightarrow \mathcal{L}(x) = 0$

$$D : C^n(\mathbb{R}) \to C(\mathbb{R}), D(x) = x'$$

$$D \circ D = D^2 \quad \underbrace{D \circ \dots \circ D}_{k \text{ times}} = D^k, \quad D^2(x) = D(D(x)) = D(x') = x''$$

$$D^k(x) = x^{(k)}$$

$$(a_1 D^k)(x) = a_1 x^{(k)}$$

$(D + a_n)(x) = x' + a_n x$, $a_1 \dots a_n \in \mathbb{R}$

$r_1, r_2 \in \mathbb{C}$ $\quad (D - r_1) \circ (D - r_2) = D' - (r_1 + r_2) D + r_1 r_2 = (D - r_2) \circ (D - r_1)$

$(D - r_1) \circ (D - r_2)(x) = (D - r_1)(x' - r_2 x)$

$= D(x' - r_2 x) - r_1(x' - r_2 x) = x'' - r_2 x' - r_1 x' + r_1 r_2 x$

$$(D - r_1) \circ (D - r_2) \overset{not}{=} (D - r_1)(D - r_2)$$

Lemma 1 : (The decomposition of $\mathcal{L}$)

Let $r_1, \dots, r_n$ be the roots of the char pol ℓ

Then $\ell(r) = (r - r_1) \dots (r - r_n)$ and $\mathcal{L} = \ell(D) = (D - r_1) \dots (D - r_n)$

Proof: $\ell(r) = r^n + a_1 r^{n-1} + \dots + a_{n-1} r + a_n$

$$\mathcal{L}(x) = x^{(n)} + a_1 x^{(n-1)} + \dots + a_{n-1} x' + a_n x$$

$$= \ell(D)(x) \Rightarrow \mathcal{L} = \ell(D)$$

ex: Let $\ell(r) = (r - 1)^2 (r + 5)$: degree 3

$$\mathcal{L} = (D - 1)^2 (D + 5)$$

Compute $\mathcal{L}(1)$ for $x \in \{e^{-t}, e^{-5t}, e^t, te^t, t^2 e^t\}$

$x = e^{-t}$

$$(D+5)(e^{-t}) = (e^{-t})' + 5 \cdot e^{-t} = 4e^{-t}$$

$$(D-1)(e^{-t}) = (e^{-t})' + (-1) \cdot e^{-t} = -2e^{-t}$$

$$\mathcal{L}(e^{-t}) = (D-1)^2 (D+5)(e^{-t}) = (D-1)^2 (4e^{-t})$$
$$= 4(D-1)(D-1)(e^{-t})$$
$$= 4(D-1)(-2e^{-t})$$
$$= -8(D-1)(e^{-t})$$
$$= -8(-2e^{-t})$$
$$= 16e^{-t}$$

$$\mathcal{L}(e^{-t}) = 16e^{-t}$$

$x = e^{-5t}$

$$(D+5)(e^{-5t}) = (e^{-5t})' + 5e^{-5t} = 0$$

$$\mathcal{L}(e^{-5t}) = (D-1)^2(0) = 0$$

$\mathcal{L}(e^{-5t}) = 0 \Rightarrow e^{-5t}$ is a sol of $\mathcal{L}(x) = 0$

$x = e^{t}$

$(D-1)(e^{t}) = 0 \Rightarrow \mathcal{L}(e^{t}) = (D+5)(D-1)^2(e^{t}) = 0 \Rightarrow$
e^{t} is a sol.

$$(D-1)(te^{t}) = e^{t}$$

$$\mathcal{L}(te^{t}) = (D+5)\underbrace{(D-1)(D-1)(te^{t})}_{0} = 0$$

- $x = t^2 e^t$

$$\mathcal{L}(t^2 e^t) = \text{const}\, e^t$$

Copy the part that should be here !!

Exercises :

1) Prove that $\{\sin t, \cos t, e^{2t}\}$ are linearly indp. using the def.

2) Prove that $\{\sin t, \cos t, e^{2t}\}$ are lin. indep using Th 1

sol'e to 2)

$e^{2t} \longmapsto r_1 = 2$ is a root of the charact. pol ℓ

$\cos t$ and $\sin t \longmapsto r_2 = i$, i a root of

$\ell(r) = (r-2)(r-i)(r+i)$

incomplete

ex : 3) we consider $\mathcal{L}$ (general)

Let x be a constant function. Then $\mathcal{L}(x)$ is also a const funct

Let $x = d e^{\alpha t}$, then $\mathcal{L}(d e^{\alpha t}) = d \underbrace{\ell(\alpha)}_{\text{const}} e^{\alpha t}$

$(D - r_1)(d e^{\alpha t}) = (d e^{\alpha t})' - r_1 d e^{\alpha t} = d(\alpha - r_1) e^{\alpha t}$

2) $\mathcal{L}_n$ - HDE with CC

Th 2: The fund. th

The general sol of $\mathcal{L}(x) = f(t)$ is

$x = x_h + x_p$, where

x_p is a particular sol of $\mathcal{L}(x) = f(t)$ and
x_h is the gen sol of $\mathcal{L}(x) = 0$

Proof x_p is a

~~1.4~~, ~~1.5~~, ~~1.6~~, ~~1.8~~, ~~1.9~~, ~~1.15~~, 1.16

• Homework

1.4. Problem : periodic response to a periodic force

$\varphi(t) = \sin 2.2t - \sin 2t$ for all $t \in \mathbb{R}$ is sol of IVP

$$x'' + 4.84x = -0.84 \sin 2t \quad x(0) = 0$$
$$x'(0) = 0.2$$

1) Verif. the diff eq :

$$x'' = \varphi(t)'' = ((\cos 2.2t) \cdot 2.2)' - (2 \cos 2t)'$$
$$= -\sin 2.2t \cdot (2.2)^2 + 4 \sin 2t$$
$$= -4.84 \cdot \sin 2.2t + 4 \sin 2t$$

$$\begin{array}{r} 2.2 \cdot \\ 2.2 \\ \hline 4,84 \end{array}$$

$$\cancel{-4,84 \sin 2,2t} + 4 \sin 2t + \cancel{4,84 \sin 2,2t} - 4,84 \sin 2t$$

$$= -0,84 \sin 2t$$ ✓ verification done

=> the diff. eq. is satisfied

2) Verif the initial conditions :

Pos at $t=0$

$\varphi(0) = \sin 0 - \sin 0 = 0$

so $x(0) = 0$

$\varphi'(0) = 2.2 \cos 0 - 2 \cos 0 = 2.2 - 2 = 0.2$

so $x'(0) = 0.2$ ✓

=> Hence $\varphi(t)$ is indeed a solution of the IVP

3) Minimal period of force

$f(t) = -0,84 \sin(2t)$

for $\sin(wt)$

$$T = \frac{2\pi}{w}$$

$$w = 2 \Rightarrow T_f = \frac{2\pi}{2} = \pi$$ the minimal period of the force

4) Minimal period of the response

Response: $\varphi(t) = \sin(2.2t) - \sin(2t)$

for $\sin(2.2t)$:

$$T_1 = \frac{2\pi}{2.2} = \frac{10\pi}{11}$$

for $\sin(2t)$:

$$T_2 = \frac{2\pi}{2} = \pi$$

$2.2T = 2\pi m \quad 2T = 2\pi m$

$T = \frac{10\pi}{11} m = \pi m$

$\frac{10}{11} m = m$

$\Rightarrow m = 11$

$m = 10$

$$T = 11 \cdot \frac{10\pi}{11} = 10\pi$$

The minimal period of the response

Perioada funcției $\sin(wt)$

$\sin(t) \mapsto$ repeats at every 2π

$\sin(t+2\pi) = \sin(t)$
so the period is 2π

If we have $\sin(wt)$

we want T: $\sin(w(t+T)) = \sin(wt)$

$\sin(wt + wT)$

$$\boxed{wT = 2\pi \Rightarrow T = \frac{2\pi}{w}}$$

Formula of the period

1.5 Problem: quasiperiodic response to a periodic force

$\varphi(t) = \sin\sqrt{6}t - \sin 2t \quad t \in \mathbb{R}$ is a sol. of the IVP

$$x'' + 6x = +2\sin 2t \qquad x(0) = 0 \qquad x'(0) = 0$$

$$x'' = \varphi''(t) = (\sqrt{6}\cos\sqrt{6}t)' - (2\cos 2t)' = 6\sin\sqrt{6}t + 4\sin 2t$$

$$-6\sin\sqrt{6}t + 4\sin 2t + 6\cancel{\sin\sqrt{6}t} - 6\sin 2t$$

$$= -2\sin 2t$$ ✓ satisfied

$\varphi(0) = \sin 0 - \sin 0 = 0$

$\varphi'(0) = \sqrt{6}\cdot 1 - 2\cdot 1 = \sqrt{6} - 2$

quasiperiodic $\Rightarrow$ the terms are periodic
but their summ doesn't have the same periodicity

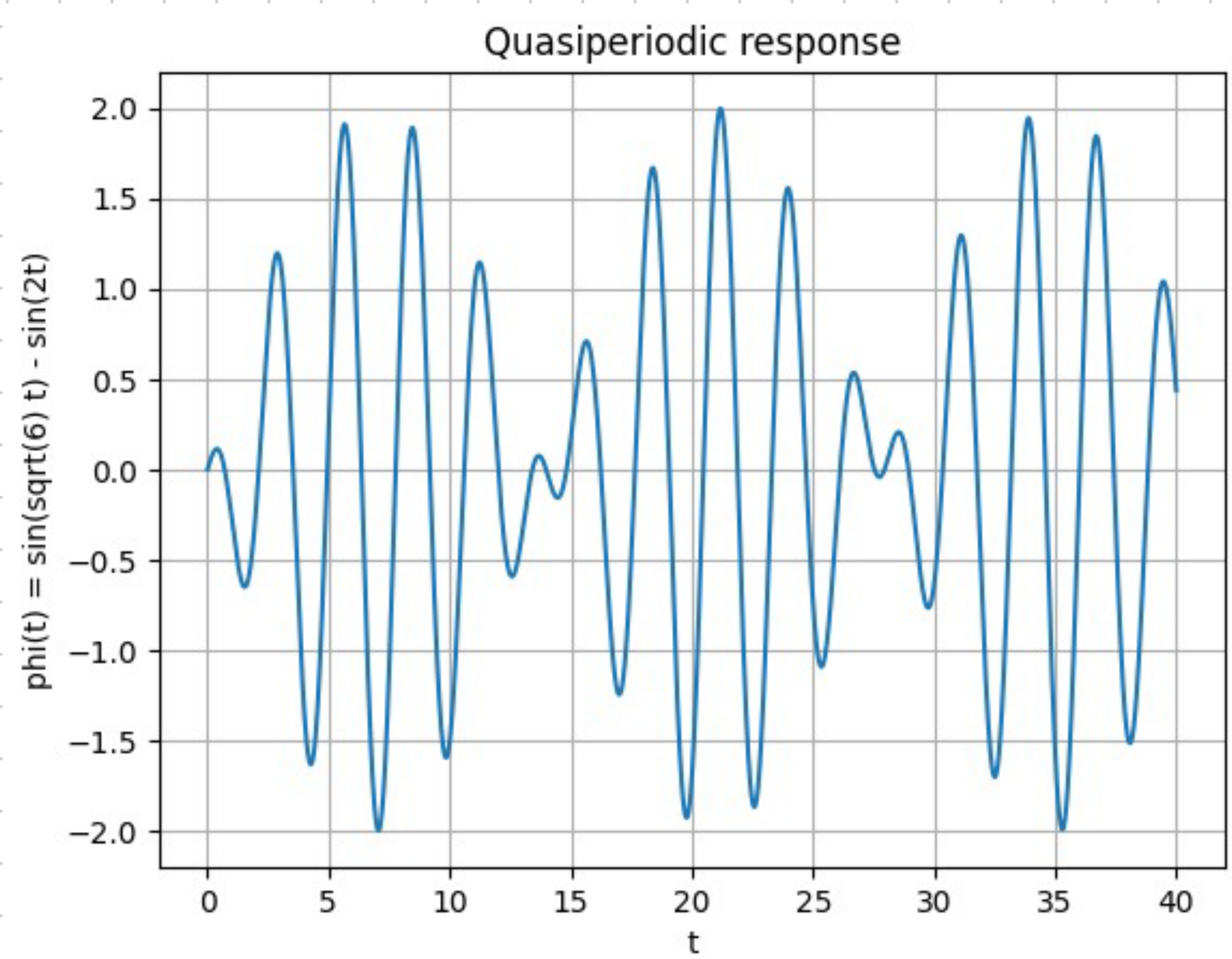
Trick: look at the frequencies ratio

$$\varphi(t) = \sin(\sqrt{6}\,t) - \sin(2t)$$

$W_1 = \sqrt{6} \quad w_2 = 2 \Rightarrow \frac{w_1}{w_2} = \frac{\sqrt{6}}{2} \notin \mathbb{Q}$

if the result $\in \mathbb{Q} \Rightarrow$ sum is periodic

↳ else the sum quasiperiodic



1.6. $\varphi(t) = t \sin t \quad t \in \mathbb{R}$

$x'' + x = 2\cos t$

$x(0) = 0 \;,\; x'(0) = 0$

$\varphi''(t) = (\sin t)' + (t \cos t)'$

$\varphi''(t) = \cos t + \cos t - t \sin t$

$\varphi''(t) = 2\cos t - t \sin t$

$\varphi''(t) + \varphi(t) = 2\cos t - \cancel{t \sin t} + \cancel{t \sin t} = 2\cos t$

$f(0) = 0 \cdot \sin 0 = 0$

$f'(0) = \sin 0 + 0 \cdot \underbrace{\cos 0}_{1} = 0$

Long-term behavior:

$t \to \infty$

$-1 \leq \sin t \leq 1 \quad | \cdot t$

$-t \leq t \sin t \leq t$

Eg: $x'' + x = 2\cos t$

The system without force

$x'' + x = 0$

↳ sol $\sin t, \cos t$

↳ natural freq $\to 1$

external force $= 2\cos t$

Solution curve:

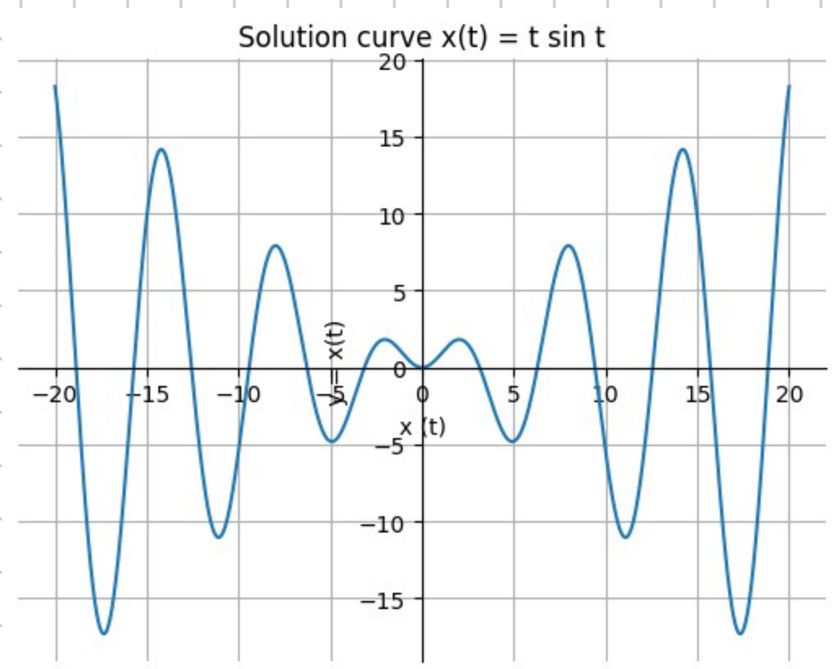
$x(t) = t \sin t$

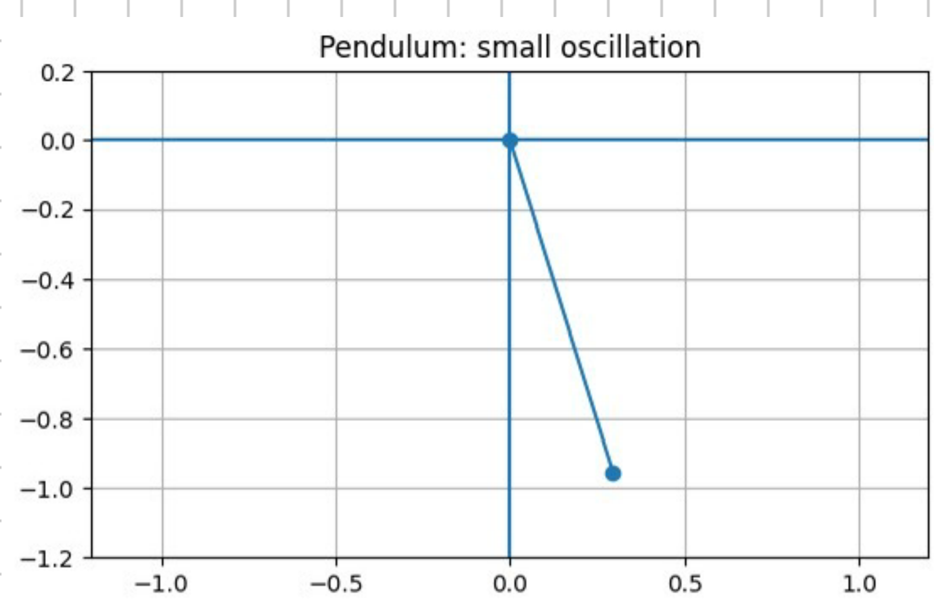
$-t \leq t \sin t \leq t$

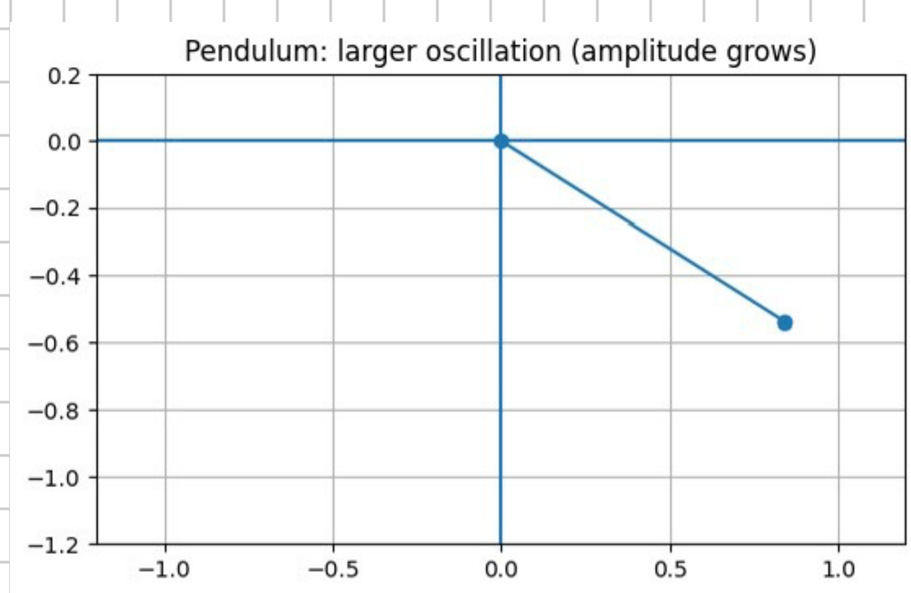
$-1 \leq \sin t \leq 1$

So the oscillation stays between two straight lines: $y = t$ and $y = -1$

The graph oscillates, but with amplitude growing like t.







1.8 a) $x' = x - x^3$

$x(t) = c$

$x'(t) = 0$

$0 = f(c)$

$0 = x - x^3$

$0 = x(1-x^2)$

$0 = x(1-x)(1+x)$

$x = 0,\ x = 1,\ x = -1$

b) $x' = \sin x$

$0 = \sin x$

$x = k\pi \quad k \in \mathbb{Z}$

Constant sol:

$x(t) = k\pi,\ k \in \mathbb{Z}$

c) $x' = \frac{x+1}{2x^2+5}$

$\frac{x+1}{2x^2+5} = 0 \qquad 2x^2+5>0$ always

$x = -1$

$x(t) = -1$

d) $x' = x^2 + x + 1$

$x^2 + x + 1 = 0$

$\Delta = 1 - 4 = -3 < 0$

No real const sols.

e) $x' = x + 4x^3$

$x + 4x^3 = 0$

$x(1+4x^2) = 0$

$x = 0$

$4x^2 = -1$

$x^2 = -\frac{1}{4}$

no real sols $\Rightarrow x(t) \equiv 0$

f) $x' = -1 + x + 4x^3$

$0 = -1 + x + 4x^3$

$4\cdot\left(\frac{1}{2}\right)^3 + \frac{1}{2} - 1 = \frac{4}{8} + \frac{1}{2} - 1$

$= \frac{1}{2} + \frac{1}{2} - 1 = 0$

so $x = \frac{1}{2}$ is a root

$(2x-1)(2x^2+x+1) = 0$

$x = \frac{1}{2}$

$2x^2 + x + 1 = 0$

$\Delta = 1 - 8 = -7 < 0$

$x(t) = \frac{1}{2}$

1.9 $x_1, x_2, x_3 : \mathbb{R} \to \mathbb{R}$

(i) $x_1(1) = 1, \quad x_2(t) = t, \quad x_3(t) = t^2, \quad \forall t \in \mathbb{R}$

linearly indep

$$a x_1(t) + b x_2(t) + c x_3(t) = 0$$

$$a = b = c = 0$$

$$a \cdot 1 + b \cdot t + c \cdot t^2 = 0$$

$a + bt + ct^2 = 0$ (A) $\forall t \in \mathbb{R}$ $\Rightarrow$ $a = 0$, $b = 0$, $c = 0$ $\Rightarrow$ x_1, x_2, x_3 linearly indep

(ii) $x' - 5x = 2t^2 + 3$

$$x'(t) = 2at + b$$

$$2at + b - 5at^2 - 5bt - 5c$$

$$-5at^2 + (2a - 5b)t + (b - 5c) = 2t^2 + 3$$

$-5a = 2 \qquad a = -\frac{2}{5}$

$b - 5c = 3 \qquad -\frac{4}{25} - 5c = 3$

$2a - 5b = 0 \qquad -5c = 3 + \frac{4}{25} \quad | \cdot 25$

$-\frac{4}{5} - 5b = 0 \qquad -125c = 75 + 4$

$-5b = \frac{4}{5} \qquad -125c = 79$

$b = -\frac{4}{25} \qquad c = -\frac{79}{125}$

$$x(t) = -\frac{2}{5}t^2 - \frac{4}{25}t - \frac{79}{125}$$

1.15

$$x = c_1 e^{at} + c_2 e^{-at}, \quad c_1, c_2 \in \mathbb{R}$$

$$x'' = a^2 x$$

$$y = x' - ax$$

$$y' = -ay$$

$$y' = x'' - ax'$$

$$y(t) = ke^{-at}$$

$$x' - ax = ke^{-at}$$

solving the non hom. ODE:

$$x(t) = -c_1 e^{at} + c_2 e^{-at}$$

Seminar 3

1.7 $f(t) = \cos t$

a) $x' + x = 0$

$f'(t) = -\sin t$

$-\sin t + \cos t = 0$

$\cos t = \sin t$

not a solution

$x^{(4)} + x'' = 0$

$x'' = -\cos t$

$x^{(4)} = \cos t$

$\cos t - \cos t = 0$

$\Rightarrow f(t)$ sol

1.11 $x(t) = e^{rt}$ $\quad r \in \mathbb{R}$

a) $x'' - 5x' + 6x = 0$

Rest - in the pink notebook

Lab 1

$$x^{(n)}(t) + \dots + a_1(t)x'(t) + a_0(t) \cdot x(t) = f(t)$$

$x'(t) = 1$

$x''(t) = 1$ acceleration

$x(0) = c_1 \qquad x'(0) = c_2$

Lecture 4

Linear diff eq. with constant coeff

$\mathcal{L}: C^n(\mathbb{R}) \to C(\mathbb{R})$, $x \in C^n(\mathbb{R}) \longmapsto \mathcal{L}(x) =$
$x^{(n)} + a_1 x^{(n-1)} + \dots + a_n x_n$ where $n \in \mathbb{N}^*$, $a_1, \dots, a_n \in \mathbb{R}$

Notation : $l(r) = r^n + a_1 r^{n-1} + \dots + a_{n-1} r + a_n$
$\Rightarrow$ the charact. polynomial of the diff op. $\mathcal{L}$.

In order to find the gen. sol of the LHDE $\mathcal{L}(x)=0$ we have the characteristic equation method.

We need $r_1, \dots, r_n \in \mathbb{C}$ the roots of the char pol.

$$l(r) = (r - r_1) \dots (r - r_n)$$

If $r_1 = \alpha \in \mathbb{R}$ is a root of mult. $m \geq 1$ $\Rightarrow$ we ~~asso~~ assoc. the sol
$\mapsto e^{\alpha t}, t e^{\alpha t}, \dots, t^{m-1} e^{\alpha t}$

If $r_{1,2} = \alpha \pm i\beta$ $(\alpha, \beta \in \mathbb{R}, \beta > 0)$ root of multiplicity $m \geq 1 \Rightarrow e^{\alpha t} \cos \beta t$, $e^{\alpha t} \sin \beta t$,
$t e^{\alpha t} \cos \beta t, \dots t^{m-1} e^{\alpha t} \sin \beta t$

(2m functions)

We obtain $x_1, \dots, x_n$ (n functions)

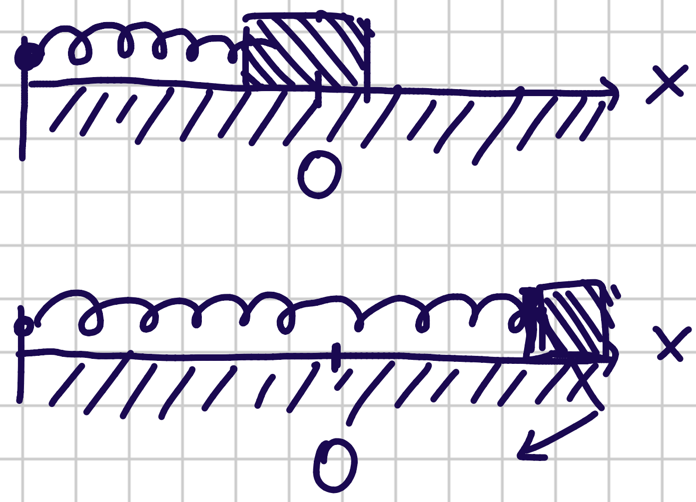
the gen. sol. of the n^{th} order LHDE $\mathcal{L}(x)=0$ is $x = c_1 x_1 + \dots + c_n x_n$, $c_n \in \mathbb{R}$

We consider now the L_n-HDE

(1) $\mathcal{L}(x) = f(t)$

Then $f \in C(\mathbb{R})$

The spring-mass system



F_r = the restoring

F_f = the friction

The second Newton's law

$$\boxed{m x''(t) = -F_r - F_f + f(t)}$$

- Hooke's law: $F_r = Kx$; $K > 0$
- $F_f = \gamma x'$; $\gamma \geq 0$ ← damping coeff

$$mx'' = -Kx - \gamma x' + f(t)$$

$$\boxed{x'' + \frac{\gamma}{m} x' + \frac{k}{m} x = f(t)}$$

Case. 1. $\gamma = 0$; $f(t) \equiv 0$

$$x'' + \frac{k}{m} x = 0$$

Un-damped motion, without external force

Case 2

$$\gamma > 0, \ f(t) \not\equiv 0$$

$$x'' + \frac{\gamma}{m} x' + \frac{K}{m} x = 0$$

Case 3

$$\gamma = 0, \ f(t) = A\cos(\omega t)$$

(bounded, periodic, external force)

$$x'' + \frac{K}{m} x = A\cos(\omega t)$$

<u>Case 1</u> $x'' + \frac{K}{m} x = 0$ $\mapsto \cos(\omega_0 t), \sin(\omega_0 t)$

$\omega_0 = \sqrt{\frac{K}{m}} > 0$

$x = c_1 \cdot \cos(\omega_0 t) + c_2 \cdot \sin(\omega_0 t)$; $c_1, c_2 \in \mathbb{R}$

$x'' + \omega_0^2 x = 0$

$r^2 + \omega_0^2 = 0$

any such funct. is $\frac{2\pi}{\omega_0}$ $\to$ periodic oscillatory with const. amplitude

$r^2 = -\omega_0^2 < 0$

$r_{1,2} = \pm i\omega_0$

For $(c_1, c_2) \in \mathbb{R}^2 \setminus \{(0,0)\}$ we take the polar coordinates

$$A_0 = \sqrt{c_1^2 + c_2^2}$$

$$c_1 = A_0 \cos\varphi_0$$

$$c_2 = A_0 \sin\varphi_0$$

$$\operatorname{tg}\varphi_0 = \frac{c_2}{c_1}$$

$$x_0 = A_0 \cos\varphi_0 \cos(w_0 t) + A_0 \sin\varphi_0 \sin(w_0 t)$$
$$= A_0 \cos(w_0 t - \varphi_0)$$

A_0 is the amplitude of the oscillation

Case 3

$$x'' + w_0^2 x = A\cos(wt)$$
$$x_h = c_1 \cos(w_0 t) + c_2 \sin(w_0 t) \; ; c_1, c_2 \in \mathbb{R}$$
$$x = x_h + x_p \; ; \; x_p = ?$$

Case 3.1

$w \neq w_0$

$$x_p = \alpha_1 \cos(wt) + \alpha_2 \sin(wt)$$

$\alpha_1, \alpha_2 - ?$

$$x = c_1 \cos(w_0 t) + c_2 \sin(w_0 t) + x_p$$

$c_1, c_2 \in \mathbb{R}$

bounded
oscillatory

Case 3.2

$w = w_0$

$\alpha_1, \alpha_2 = ?$

$$x_p = t(\alpha_1 \cos(w_0 t) + \alpha_2 \sin(w_0 t))$$

Seminar 3

The integrating factor method to find the general sol of

$$\boxed{x'(t)+a(t)\cdot x(t)=f(t)} \quad / \cdot \mu(t) = e^{-A(t)}$$

We choose : $\mu(t)=e^{-A(t)}=e^{+\int_{t_0}^{t} a(s)ds}$ is an integrating factor

$$\underbrace{x'(t)\cdot e^{-A(t)}+a(t)\cdot x(t)\cdot e^{-A(t)}}_{\left(x(t)\cdot e^{-A(t)}\right)'} = f(t)\cdot e^{-A(t)}$$

$$\left(x(t)\cdot e^{-A(t)}\right)' = f(t)\cdot e^{-A(t)} \quad \Big/ \int_{t_0}^{t} ds$$

$$x(t)\cdot e^{-A(t)} = \int_{t_0}^{t} f(s)\cdot e^{-A(s)}\,ds + c \quad \Big| \cdot e^{A(t)}$$

$$x(t) = C\cdot e^{A(t)} + e^{A(t)}\int_{t_0}^{t} f(s)\cdot e^{-A(s)}\,ds, \quad \forall c\in\mathbb{R}$$

Example:

$$x' - 2x = 4 - t \quad 1 \cdot e^{-2t}$$

$$x(t) = x_e(t) + x_p(t)$$

$$\mu(t) = e^{\int -2 dt} = e^{-2t}$$

$$\underbrace{x' \cdot e^{-2t} - 2x \cdot e^{-2t}} = (4-t) \cdot e^{-2t}$$

$$(x \cdot e^{-2t})' = (4-t) \cdot e^{-2t} \quad / \int dt$$

$$x \cdot e^{-2t} = \int (4-t) \cdot e^{-2t} dt \Rightarrow \quad u = 4-t$$

$$x(t) = -2 + \frac{t}{2} + \frac{1}{t} + c e^{2t}$$

$$v = e^{-2t}$$

$$v' = -\frac{1}{2} \cdot e^{-2t}$$

$$I = -\frac{1}{2} \int (4-t)(e^{-2t})' dt$$

$$I = -\frac{1}{2} \cdot (4-t) \cdot e^{-2t} + \frac{1}{2} \int -e^{-2t} dt$$

$$I = -\frac{1}{2} \cdot (4-t) \cdot e^{-2t} + \frac{1}{4} \cdot e^{-2t} + c$$

$$I = e^{-2t} [-2 + \frac{t}{2} + \frac{1}{4}] + c$$

$$x(t) \cdot e^{-2t} = e^{-2t} \cdot (-2 + \frac{t}{2} + \frac{1}{4}) + c \;/ e^{2t}$$

$$x(t) = -2 + \frac{t}{2} + \frac{1}{4} + c e^{2t}$$

Linear Homog. Differential Eq. with constant coefficients

4.1 a) $x''' = 0$

b) $x'''' - x =$

i) I Alg

$x''' = 0$

$\Leftrightarrow r^3 = 0 \Rightarrow 3 < d \quad r_1 = 0 = r_2 = r_3$

b) $x_1(t) = e^{r_1 t} = 1$

$x_2(t) = t \cdot e^{r_1 t} = t$

$x_3(t) = t^2 e^{r_1 t} = t^2$

$\Rightarrow x(t) = c_1 + c_2 t + c_3 t^2$

$c_1, c_2, c_3 \in \mathbb{R}$

II Integr

$x''' = 0 \,/\int$

$\int x''' dx = \int 0 \, dt$

$x'' = 0 + c_1 \,/\int$

$\int x'' dx = \int c_1 dt$

$\Leftrightarrow x' = c_1 t + c_2 \,/\int$

$x(t) = c_1 \frac{t^2}{2} + c_2 t + c_3$

e) $x''' - x = 0 \Leftrightarrow r^4 - 1 = 0 \Leftrightarrow (r^2 - 1)(r^2 + 1) = 0$

$\Rightarrow (r-1)(r+1)(r^2+1) = 0$

$x_1(t) = e^{r_1 t}$

$x_2(t) = e^{r_2 t}$

$x_3(t) = e^{r_3 t} \cos(3t)$

$x_4(t) = e^{\alpha t} \sin(\beta t)$

$r_1 = 1$

$r_2 = -1$

$r_3 = -i$

$r_4 = i$

$x(t) = c_1 \cdot e^{t} + c_2 \cdot e^{-t} + \cos(t) \cdot c_3 + c_4 \cdot \sin(t)$

4.2 a) e^{-3t} and e^{5t}

$r_1 = -3, \; r_2 = 5$

$(r - r_1)(r - r_2) = 0$

$(r+3)(r-5) = 0$

$r^2 - 5r + 3r - 15 = 0$

$r^2 - 2r - 15 = 0$

$x'' - 2x' - 15x = 0$

$x = c_1 e^{-3t} + c_2 \cdot e^{5t}$

b) $5e^{-3t}$ and $-3e^{5t}$

$\Rightarrow r_1 = -3$

$r_2 = 5$

$x'' - 2x' - 15x = 0$

$x = c_1 \cdot 5 \cdot e^{-3t} + c_2 \cdot 3 e^{5t}$

c) $5e^{-3t} - 3e^{5t}$

same conclusion

d) $5te^{-3t}$ and $-3e^{5t}$

$\Rightarrow r_1 = r_2 = r_3$

$r_3 = 5$

$x_1 = e^{r_1 t}$ $\quad r_1 = 5$

$x_2 = e^{r_2 t}$ $\quad r_2 = -3$

$x_2 = t e^{r_2 t}$ $\quad r_2 = -3$

$(r+3)^2(r-5)=0$

$(r^2+6r+9)(r-5)=0$

$r^3-5r^2+6r^2-30r+9r-45=0$

$r^3+r^2-21r-45=0$

$\Rightarrow x'''+x''-21x'-45x=0$

$\Rightarrow x=c_1\cdot e^{5t}+c_2e^{-3t}+c_3t\cdot e^{-3t}$

f) $(5-3t)e^{-3t}$

$r_1=r_2=3$

$(r+3)^2=0$

$r^2+6r+9=0$

$x''+6x'+9x=0$

$x=c_1e^{-3t}+c_2t\cdot e^{-3t}$

Find sol for each of the f. IVPs

4.4. a) $x'=3x$, $x(0)=2$

b) $x''+x=0$ $x(0)=0$, $x'(0)=\frac{1}{6}$

a) $x'-3x=0$

I $r-3=0\Rightarrow r=3$

$x(t)=c_1e^{3t}$, $x(0)=2$ $\Big|\Rightarrow c_1=2\Rightarrow x(t)=2e^{3t}$

II $\mu(t)=e^{\int -3\,dt}=e^{-3t}$

$$(=) \ (x' - 3x) e^{-3t} = 0$$

$$x' e^{-3t} - 3 e^{-3t} x = 0$$

$$(=) \ (x \cdot e^{-3t})' = 0 \ / \int$$

$$=) \ x \cdot e^{-3t} = C$$

$$x(t) = C \cdot e^{3t}$$

$$f(0) = 2 \qquad =) C = 2$$

$$x(t) = 2e^{3t}$$

b) $x'' + x = 0$

$$r^2 + r = 0$$

$$r = \pm i$$

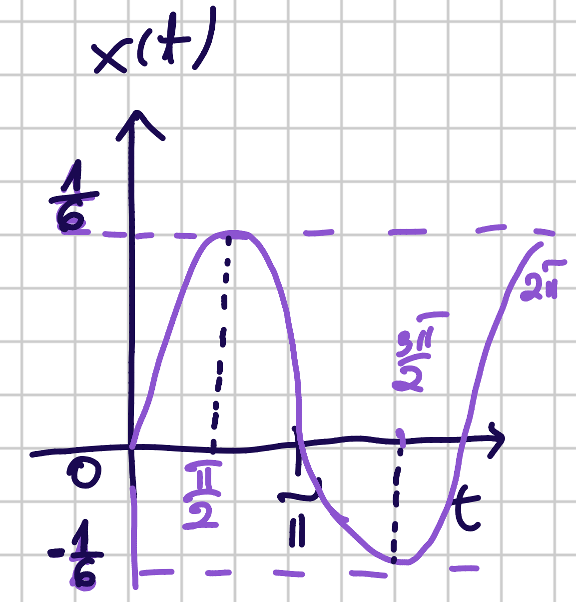
$$(=) \ x(t) = \cos(t) + \sin(t) c_2$$

$$x(0) = 0 =) \ c_1 = 0$$

$$x'(t) = c_1 \sin t + c_2 \cos(t)$$

$$x'(0) = \frac{1}{6} =) \ c_2 = \frac{1}{6}$$

$$=) \ x(t) = \frac{1}{6} \cdot \sin(t)$$



x(t)=0 ∈ S

4.5 BVP

a) $x'' + x = 0, x(0) = x(\pi) = 0$

$$r^2 + r = 0$$

$$x = c_1 \cos t + c_2 \sin t$$

$$x =$$

$$\begin{cases} c_1 = 0 \\ -c_1 = 0 \end{cases} =) \ x(t) = c_2 \sin t, \ c_2 \in \mathbb{R}$$

5.2

Decide whether the foll. stat. are T or F?

a) all the sol of

$x'' + 3x' + x = 1$

satisfy $\lim_{t\to\infty} x(t) = 1$

$x(t) = x_h(t) + x_p(t)$

x_h: the general sol of the homogenous eq.

x_p: particulary sol. of non.hom. equation

$x'' + 3x' + x = 0$

$\Leftrightarrow r^2 + 3r + 1 = 0$

$\Delta = 9 - 4 = 5$

$\Rightarrow r_{1,2} = \frac{-3 \pm \sqrt{5}}{2}$

$\Rightarrow x_h(t) = c_1 \cdot e^{\frac{-3+\sqrt{5}}{2}} + c_2 \cdot e^{-\frac{3-\sqrt{5}}{2}}$

$a'' + 3a' + a = 1 \Rightarrow a = 1$

$\Rightarrow x(t) = c_1 \cdot e^{\frac{-3+\sqrt{5}}{2}} + c_2 \cdot e^{\frac{-3-\sqrt{5}}{2}} + 1$

$\frac{-3+\sqrt{5}}{2} < 0 \Rightarrow \lim_{t\to\infty} e^{\frac{-3+\sqrt{5}}{2}t} = 0$

$-\frac{3-\sqrt{5}}{2} < 0 \Rightarrow \lim_{t\to\infty} e^{-\frac{3-\sqrt{5}}{2}t} = 0$

$\Rightarrow \lim_{t\to\infty} x(t) = 1$

b) the sol of the IVP

$x'' + 4x = 1$

$x(0) = \frac{5}{4}$, $x'(0) = 0$

satisfies $x(\pi) = \frac{5}{4}$?

$x'' + 4x = 0$

$r^2 + 4 = 0 \Rightarrow r_{1,2} = \pm 2i$

$x_1 = \cos(2t)$

$x_2 = \sin(2t)$

$x_h(t) = c_1 \cos(2t) + c_2 \sin(2t)$

$x_p(t) = a$

$a'' + 4a = 1 \Rightarrow a = \frac{1}{4}$

$x(t) = c_1 \cdot \cos(2t) + c_2 \cdot \sin(2t) + \frac{1}{4}$

$x(0) = \frac{5}{4} \Rightarrow c_1 + \frac{1}{4} = \frac{5}{4}$

$c_1 = 1$

$x'(t) = -2c_1 \cdot \sin(2t) + 2 \cdot c_2 \cdot \cos(2t)$

$x'(0) = 2c_2 = 0 \Rightarrow c_2 = 0$

$x(t) = \cos(t) + \frac{1}{4}$

$x(\pi) = 1 + \frac{1}{4} = \frac{5}{4}$ True

c) $x' = 3x + t^3$

$x_h(t) + x_p(t) = x(t)$

Homework

↳ ex. we didn't do